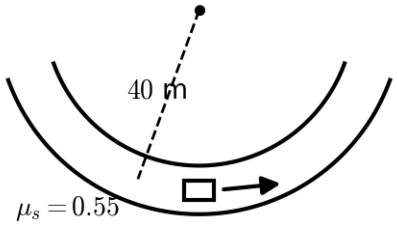
PHYS1408 Practice Exam 2-6

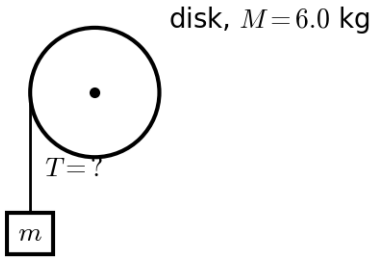
Chapters 6, 7/8, 9, 10, 11

Use $g = 9.8$ m/s^2. Choose the single best answer. Answers are rounded to the precision shown. For a minimum or maximum, choose the answer choice that actually works.

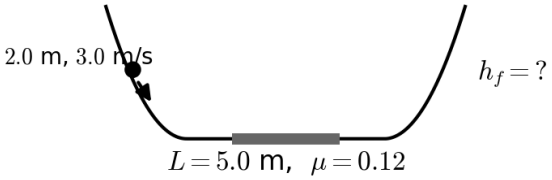
1. A car drives around a flat (unbanked) curve of radius 40 m. The coefficient of static friction between the tires and the road is $\mu_s = 0.55$. What is the maximum speed at which the car can take the curve without sliding?
a) 4.7 m/s b) 10.4 m/s c) 14 m/s d) 15 m/s
e) 19.8 m/s



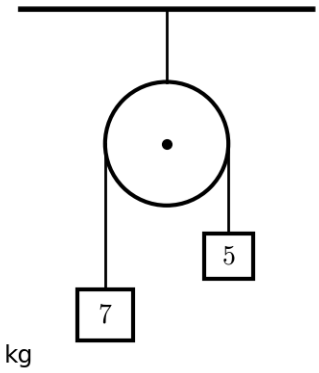
2. A 5.0 kg mass hangs from a light string wrapped around a pulley that is a uniform disk ($I = \frac{1}{2}MR^2$, $M = 6.0$ kg) on a frictionless axle. The mass is released. What is the tension in the string while the mass falls?
a) 0 N b) 18.4 N c) 26.7 N d) 30.6 N e) 49.0 N



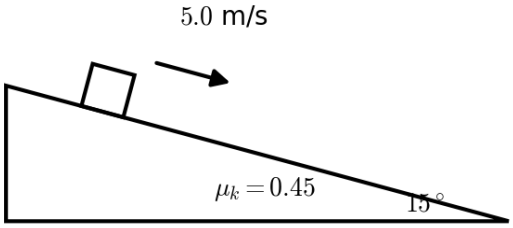
3. A skateboarder is 2.0 m up one wall of a halfpipe, moving **down** the wall at 3.0 m/s. The halfpipe is frictionless except for a 5.0 m long section of the flat bottom with $\mu = 0.12$. How high up the far wall does she rise?
a) 0.46 m b) 1.26 m c) 1.40 m d) 1.86 m e) 2.46 m



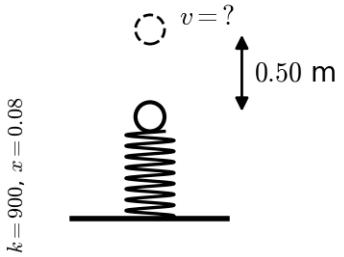
4. A 7.0 kg mass and a 5.0 kg mass hang from opposite ends of a light string over a massless, frictionless pulley. What is the magnitude of their acceleration?
a) 1.63 m/s² b) 2.80 m/s² c) 3.92 m/s² d) 9.80 m/s²
e) 19.6 m/s²



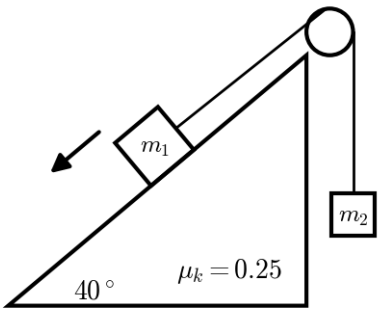
5. A block is sliding **down** a 15° incline ($\mu_k = 0.45$) at 5.0 m/s. Friction slows it down. How long does it take to come to rest?
a) 0.74 s b) 1.45 s c) 2.67 s d) 2.90 s e) it never stops; it speeds up



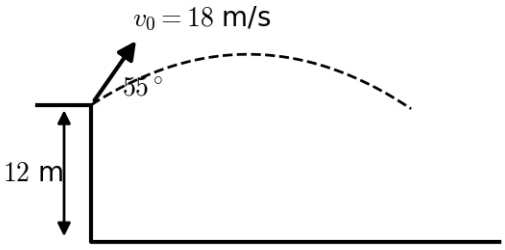
6. A vertical spring (k = 900 N/m) is compressed 0.080 m with a 0.15 kg ball resting on it, then released. How fast is the ball moving when it is 0.50 m above the release point?
a) 3.13 m/s b) 3.78 m/s c) 5.35 m/s d) 6.20 m/s
e) 8.19 m/s



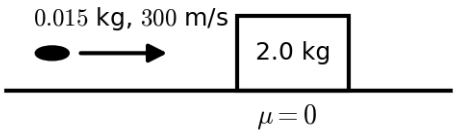
7. A 12 kg block on a 40° incline ($\mu_k = 0.25$) is tied by a light string over a massless pulley at the top to a 3.0 kg hanging mass. The block is sliding **down** the incline at 1.0 m/s (so the hanging mass is rising). How fast is the block moving after it has slid 0.80 m down the incline?
a) 1.59 m/s b) 1.88 m/s c) 2.04 m/s d) 2.43 m/s
e) it stops before it slides the whole 0.80 m



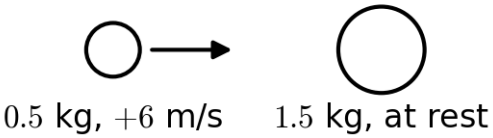
8. A ball is thrown from the edge of a 12 m high ledge at 18 m/s, 55° above the horizontal. How high above the ground below is the ball at the highest point of its path?
a) 5.4 m b) 11.1 m c) 12.0 m d) 17.4 m e) 23.1 m



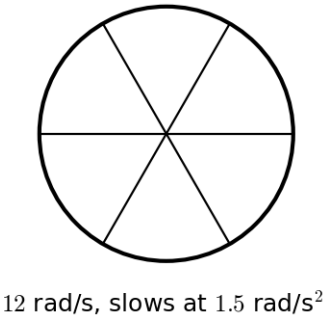
9. A 0.015 kg bullet moving at 300 m/s embeds itself in a 2.0 kg block resting on a frictionless table. How fast does the block (with the bullet) move just after the collision?
a) 2.23 m/s b) 2.25 m/s c) 4.5 m/s d) 25.9 m/s
e) 300 m/s



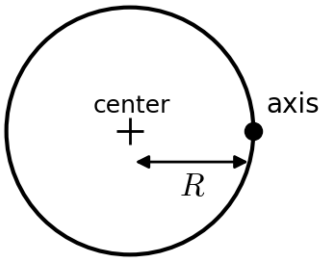
10. A 0.50 kg ball moving right at 6.0 m/s collides **elastically** head-on with a 1.5 kg ball at rest. What is the velocity of the 0.50 kg ball after the collision? Positive to the right / negative to the left.
a) +6.0 m/s b) +3.0 m/s c) +1.5 m/s d) 0 m/s
e) −3.0 m/s



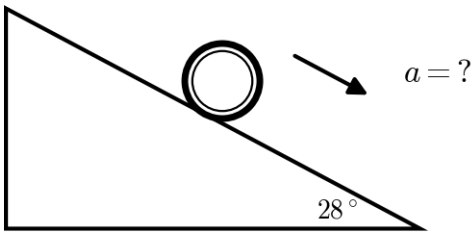
11. A wheel spinning at 12 rad/s slows down with a constant angular deceleration of 1.5 rad/s². Through what angle does it turn before it stops?
a) 7.6 rad b) 8.0 rad c) 24 rad d) 48 rad e) 96 rad



12. A uniform disk ($I_{cm} = \frac{1}{2}MR^2$, $M = 4.0$ kg, $R = 0.30$ m) rotates about an axis perpendicular to the disk through a point on its **edge**. What is its moment of inertia about this axis?
a) 0.12 kg · m² b) 0.18 kg · m² c) 0.27 kg · m²
d) 0.36 kg · m² e) 0.54 kg · m²

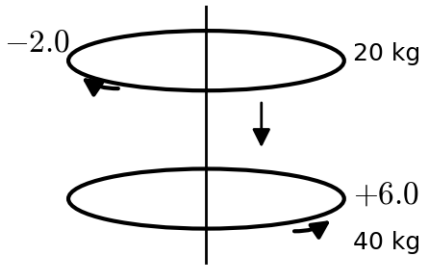


13. A thin hoop ($I = MR^2$) rolls without slipping down a 28° incline. What is the acceleration of its center?
a) 2.30 m/s² b) 3.07 m/s² c) 3.29 m/s² d) 4.33 m/s²
e) 4.60 m/s²



14. A disk ($I = \frac{1}{2}MR^2$, $M = 40$ kg, $R = 0.50$ m) spins at +6.0 rad/s. A second disk ($I = \frac{1}{2}MR^2$, $M = 20$ kg, $R = 0.50$ m) spinning the **opposite** way at −2.0 rad/s is dropped onto it, and the two end up turning together. What is their final angular velocity?

a) +2.00 rad/s b) +3.33 rad/s c) +4.00 rad/s
d) +4.67 rad/s e) +6.00 rad/s



15. A 75 kg person stands at the rim of a merry-go-round (a disk, $I = \frac{1}{2}MR^2$, $M = 300$ kg, $R = 3.0$ m) that rotates at 1.0 rad/s. The person walks **halfway** in, to 1.5 m from the center. What is the new angular velocity? (Treat the person as a point mass.)

a) 0.75 rad/s b) 1.00 rad/s c) 1.33 rad/s d) 1.50 rad/s
e) 2.00 rad/s

